

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2484

Kokomo Highland Park 138 kV
December 18, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2484. The Interconnection Customer proposes a 150.0 MW battery energy storage facility interconnecting at the Kokomo Highland Park 138 kV substation in the Pennsylvania area. The request was received on December 06, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 423 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$36,346,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
KOKOMO HIGHLAND PARK – MCLAUGHLIN 138 kV Line	IPL FIT MONROVIA I LLC – NEW JAMESTOWN 138 kV line	119.4	195	100.0
FOSTER WHEELER MT CARMEL COGEN 500 kV/345 kV XFMR	WESTWOOD GENERATION LLC 500/1kV Auto	104.9	918	5.3
WESTWOOD GENERATION LLC 345 kV/1 kV TX	FOSTER WHEELER MT CARMEL COGEN 500 kV/345 kV Autotransformer	104.8	1,170	6.9
TENNESSEE AVE – CLOVE LAKES 138 kV	MCLAUGHLIN SUBSTATION – KOKOMO HIGHLAND PARK 138 kV	103.4	170	100.0
LIGHTHOUSE HILL – AES BEAVER VLY PARTNERS BEAVER VLY SUBSTATION 345 kV	HILTOP – PINEY CRK PROJECT 500 kV	102.0	1,880	5.0
TIDD SUBSTATION – LERDO 500 kV	FILER – TIDD 500 kV L	101.0	3,129	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KOKOMO HIGHLAND PARK – MCLAUGHLIN 138 kV Line	Rebuild line with higher-capacity conductor	\$14,779,000	100% (\$14,779,000)
FOSTER WHEELER MT CARMEL COGEN 500-345 kV XFMR	Replace transformer with higher-rated unit	\$17,316,000	100% (\$17,316,000)
WESTWOOD GENERATION LLC 345 kV/1 kV TX	Replace transformer with higher-rated unit	\$13,952,000	15% (\$2,141,000)
TENNESSEE AVE – CLOVE LAKES 138 kV	Reconductor line	\$1,705,000	100% (\$1,705,000)
LIGHTHOUSE HILL – AES BEAVER VLY PARTNERS BEAVER VLY SUBSTATION 345 kV	Reconductor line	\$5,332,000	1% (\$66,000)
TIDD – LERDO 500 kV line	Reconductor line	\$28,379,000	1% (\$339,000)
	Total allocated to project		\$36,346,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.